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Preparation Methodology, Properties, and Applications of Nano-Silica in the Cement Slurry

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Abstract

Nano-silica can be produced by tuning the morphological features by controlling the synthetic conditions, such as time, temperature, and concentration of materials. This paper will describe the development methodology and laboratory testing performed to qualify an advanced nano-silica slurry recipe to develop an effective cement lost circulation plug.

The lab test was conducted following API 10B2 standards. Nano-silica was prepared by selecting conventional routes, including Stober's, sol-gel, flame synthesis, and microemulsion. The reaction mechanism of Nano silica combined with class G-cement was observed during cement hydration. Both nano silica and calcium hydroxide were dissolved, and Calcium Silicate & Hydrite (CSH) gel was formed. Later, a comparative study was conducted on the performance efficiency of the cement slurry at a typical density of 118 pcf (15.8lbm/gal). This study also focused on slurry stability, free fluid reduction, lower fluid loss value, and fine rheology.

The study proved that the cement slurry with nano-silica has shown 40% improvement in compressive strength due to the increased rate of hydration, thus thickening time was only within 5 hours compared to 7-8 hours as an average thickening time in legacy Cement Lost Circulations designs. The new CSH binder enhances the cement paste density, reinforcing the structure between cement grains. Cement slurry was designed with 118 pcf densities; a compressive strength of 50 psi was achieved at 9:35 hours: minutes, 500 psi compressive strength was achieved at 11:14 hours: minutes, and a compressive strength of 2809 psi was recorded in the 24-hour chart. Furthermore, experiments proved that the new slurry can be used for thixotropic properties to achieve more effective cementing squeeze, treating lost circulation, and fluid-loss control. In summary, this slurry design was found to be more cost-effective as well as more efficient in lost circulation prevention and placement of kick-off plugs.

This paper introduces to the industry a newly developed advanced slurry that significantly enhances the lost circulation treatment efficiency while reducing its cost at the same time. The paper will provide a detailed description of the application of Nano Silica Slurries in well cementing and lost circulation plugs. The paper will offer a discussion of the methodology, material properties, and applications.

Introduction

Due to its affordability, abundance, and promising outcomes, nano-silica (NS) is the most extensively utilized nanomaterial in the oil and gas sectors for various applications. NS is an efficient substitute for traditional additives such as calcium chloride and silica since it requires far less than others. Nano-silica functions as a multipurpose additive, and reinforcement of NS enables appropriate cementation and better durability. NS further aids in reducing the wait on cement (WOC) period and decreases the total principal price (Akshar Thakkar et al., 2020). NS is strongly suggested for offshore deep wells in which high pressure and temperatures are expected. NS has frequently been manufactured using various approaches differing in raw precursor materials, energy consumption, and synthetic conditions, including temperature, pH, precipitation time, nature of coagulant, and cleaning and drying procedure. These parameters impact the size, aggregation, and specific surface area of the NS (Shivshambhu K. et al., 2022).

Modifying sodium silicate solutions with acid and flame hydrolysis is now the most significant commercial procedure in synthesizing NS. Due to the cost of the raw ingredients and the amount of energy needed, both procedures are pricey. Nanosilica, also known as silicon dioxide nanoparticles or silica nanoparticles, is a compound of silicon and oxygen. Its chemical formula is SiO_2 , which is present in the form of white powder and Liquid form (Mardiana O. et al., 2017). Nano silica's uses may be expanded further when a new industrial, low-cost manufacturing technique is created (Askaruly K. et al., 2020). In general, nanomaterials can be synthesized in two broader strategies, including the top-down approach (from large to small) and the bottom-up approach (from small to large) (Figure 1). The top-down strategy is based on the devastation or size decrease of the concept, i.e., producing the materials from more extensive to smaller. In contrast, the bottom-up route is utilized to gather the smaller components to produce larger materials. In addition, NS can also be prepared based on the three procedures, such as physical, chemical, and biological methods (Faheem A. et al., 2022). NS can increase the density of concrete and improve the mechanical properties of concrete (Zhang et al., 1991). It promotes the pozzolanic reaction, resulting in the enhancement of durability and mechanical properties of cement; utilizing Nanotechnology in the preparation of cement slurry can also achieve solutions to some of the problems about oil well cementation (Singh, L. P., Karade L. et al., 2013). In the experimental study on the designed cement systems, it can be concluded that compositions with the addition of "Microblock" develop an increased compressive strength, less fluid loss, and low free water content compared to ordinary Portland cement (Gaurina-medimurec et al., 2017).

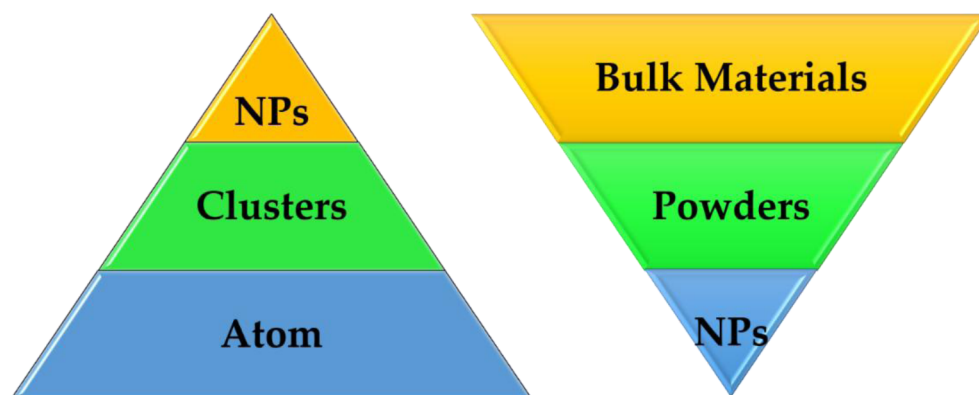


Figure 1—Strategies for synthesizing NP particles: (left to right) Bottom-up and Top-down approaches.

Further study is required to determine the optimum application of a dispersant, retarders, and the addition of extra water to the compositions containing "Microblock" for the particular well application (Patchen, F. D., 1960). It also imparts thixotropy to the slurry and can help control lost circulation and gas migration. Nano silica-based coatings have fireproof properties, and magnesium hydroxide nanoparticles, together with micro-trihydrated alumina, act as flame-retardant pigments (Sophia L. Bergen et al., 2022). The oil and

gas sector, civil engineering, and construction all benefit from using cement slurries with nano-silica added, since it is a very effective pozzolanic material comprised of extremely fine silica (Elaheh A. et al., 2018). Nano-silica comprises vitreous particles one thousand times smaller than the average cement particle size (J.A. Abdalla et al., 2022). A comprehensive review and findings on earlier research studies and current trends in concrete containing nanomaterials, including nano-SiO₂, nano-Al₂O₃, graphene oxide, carbon nanotubes, nano-TiO₂, nano-Fe₂O₃, nano-clay/metakaolin, and nano-CaCO₃ on cement concrete's water absorption and chloride penetration. Based on the analysis of various studies, the following conclusions are drawn (S. Rai et al., 2018). In this work, the compressive, flexural, and split tensile strength and coefficient of water absorption of high-performance self-compacting concrete containing different amounts of Fe₂O₃ nanoparticles have been investigated (L. Senff et al., 2009). The Fe₂O₃ nanoparticles can act as nanofillers and recover the pore structure of the specimens by decreasing harmful pores to improve water permeability (B.S. Thomas et al., 2021). Nano-CaCO₃ densifies the microstructure and enhances the concrete density by refining the pores (M.S.M. Norhasri et al., 2017). 65–70% reduction in the water absorption with 3% Nano-CaCO₃ (A.M. Ibrahim et al., 2013). Any further increase in Nano-CaCO₃ quantity in the mix led to a steep rise in water absorption capacity (J.A. Abdalla et al., 2022). It was also seen that the Nano-CaCO₃ performed the best in reducing water absorption of mixes when compared to nano-SnO₂ and nano-ZrO₂ (Selvaranjan K. et al., 2020). This significant rise in performance was deemed to be because of Nano-CaCO₃'s ability to modify the hydration reaction, where the cementitious compounds formed were of smaller crystal size (A. Khoshakhlagh et al., 2012; S.C. Devi et al., 2020).

Materials and Methodology

Synthesis of NS using chemical methods

In general, nano-silica can be produced by tuning morphological features by controlling the synthetic conditions, such as time, temperature, and concentration of precursors (N.R. Radhip et al., 2015). The chemical procedure is mainly based on the bottom-up strategy for preparing NS (Drabkaasaams V. B., 2005). In the past, NS was prepared by selecting conventional routes, including Stober's method, sol-gel, flame synthesis, and microemulsion (Liou T. H. et al., 2011). The chemical methods mentioned are simple and can have optimized parameters, but they can be expensive and challenging (Anu Mary Ealia et al., 2017; Cha J. N. et al., 2000). For instance, in the case of reverse micro-emulsion, the spherical-shaped micelles are produced by dissolving surfactants in water. Though this process is efficient, until now, it has been expensive and complicated to isolate the surfactants from the end products (Narayan R. et al., 2018). Chemical vapor condensation (CVC) is another typical process to prepare NS particles, in which silicon tetrachloride chemically interacts with hydrogen and oxygen (Zhao S. et al., 2020). The physical characteristics of the NS particles, including surface morphology and particle dimensions, can also be altered to the favorite features during the CVC route. Further, this CVC method is broadly employed to synthesize nanomaterials in powder form.

In the same way, the sol-gel process (SGP) is a well-recognized method broadly employed to produce nano silica particles and silica gel (Goodman B. A., 2020). During SGP, metal alkoxides such as TEOS or inorganic salts like sodium silicate undergo the two main reactions, hydrolysis and condensation, in the presence of an acid or base catalyst (Sahar Porrang et al., 2022). Figure 2 illustrates the most common methods to prepare NS particles (Lieberman A. et al., 2014). It is important to note that the precursors and methods presented in Figure 2 are considered the most common. However, several researchers have substituted them for desired applications (A. Mourhly et al., 2015). The hydrolysis of the metal alkoxides used to produce silica particles results in silanol groups, which polymerize to produce the silica structure (Salavati-Niasari M. et al., 2013). An alternative technique, termed Stober's process, was first investigated and discovered by Werner Stober. He prepared spherical silica particles from a silica alkoxide alcohol solution by selecting ammonia as a catalyst (Xu J. et al., 2021). This technique has improved with time and

has become much more adaptable in altering parameters and achieving the necessary features (Jafariesfad, N. et al, 2017).

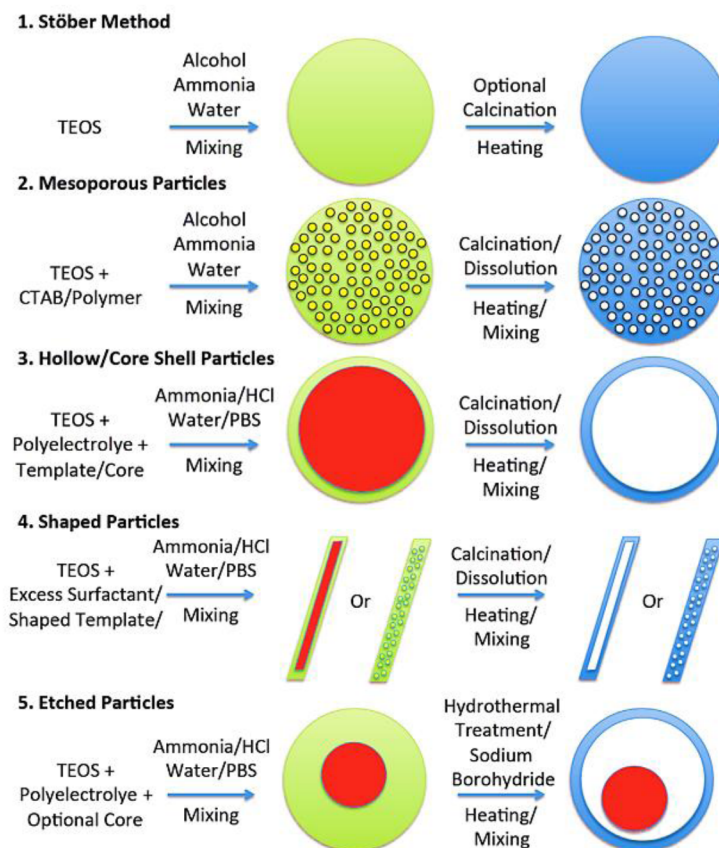


Figure 2—Frequent methods to synthesize NS particles

For instance, Nozawa et al (K. Nozawa et al., 2004) evaluated the influence of the added quantity of precursors, tetraethyl orthosilicate (TEOS), on the synthesized size of NS particles. Their results revealed that the size of the NS particles was reduced by increasing the incorporation of the TEOS and precisely altering the rate of TEOS, in addition to forming homogeneous NS particles (A. Lazaro et al., 2013). As displayed in Fig.3a–c, an increase in the rate of added TEOS from 0.005 ml/min (Fig. 1a) to 0.05 ml/min caused a 30% reduction in NS particle size (Fig. 1b). An additional rise in the rate of addition to 0.5 ml/min provided the 60% reduction in particle size (Fig. 1c). The results obtained confirmed that the higher TEOS addition perhaps offers a more density of nucleation centers depending on small particles (Bakar R. A. et al., 2016). Generally, silica gel structure is obtained using an acid catalyst, while monodispersed NS particles are prepared using the customer's method (Jeelani P. G. et al., 2020). Jafari and Allahdi prepared NS from fumed silica through a new route based on an acid-base precipitation technique (V. Jafari et al., 2014).

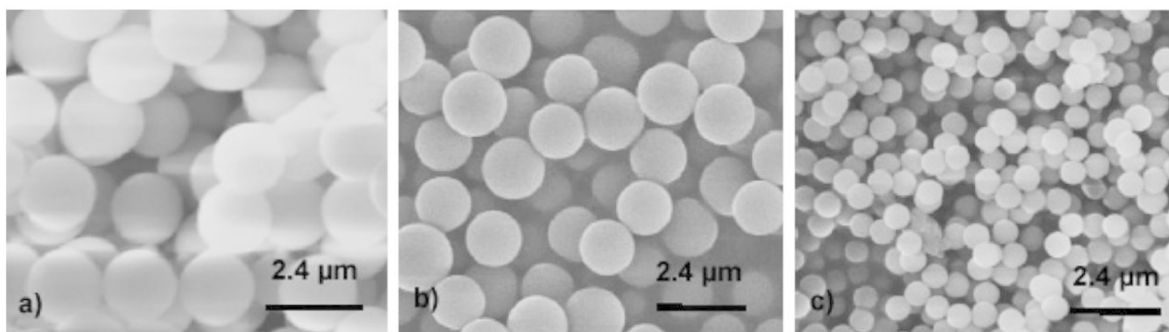


Figure 3—Surface morphological images of synthesized NS particles by varying rates of TEOS additions: (a) 0.005 ml/min, (b) 0.05 ml/min and (c) 0.5 ml/min

Results

Ultrasonic Cement Analyzer

Ultrasonic cement analyzer (UCA) equipment is shown in Figure 4. This machine is used to find out compressive strength and slurry transition time, and it is a non-destructive method to determine the slurry compressive strength development at the bottom hole static temperature and pressure parameters.



Figure 4—Ultra-Sonic Cement Analyzer (UCA)

The hypothesis of activity depends on the relationship between the ultrasonic signal speed and its compressive strength. Quality signs are dictated by estimating the adjustment in speed of an ultrasonic sign transmitted through the slurry test as it fixes. Using this device provides a slurry compressive strength chart; the UCA chart is shown in Figure 5. UCA chart plots are divided into transit time ($\mu\text{sec}/\text{min}$), temperature ($^{\circ}\text{C}$), and compressive strength (psi). Cement slurry was tested at the bottom hole static temperature. The ultrasonic waves penetrate the slurry, and the transit time peak starts from 16-18 $\mu\text{sec}/\text{min}$; the slurry design decides the transit time value. Cement slurry after 24 hours of curing in a bottom hole static temperature and bottom hole pressure of 3000 psi needs to be maintained in the machine, then compressive strength gradually increases, and the transit time value slowly decreases, as shown in the graph (Figs. 5-7).

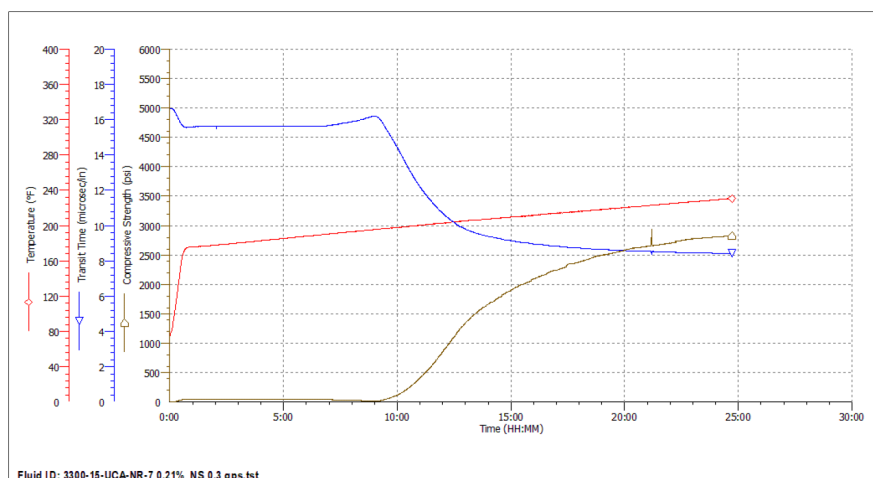


Figure 5—UCA Chart with 0.1 % DFR-1, 0.21 % R-7, 0.3 gal/sk nano-silica Cement slurry designed for 118 pcf densities, compressive strength 50 psi at 9:35 hours: minutes, 500 psi at 11:14 hours: minutes, 2809 psi compressive strength recorded in 24 hours chart.



Figure 6—Thickening Time Consistometer

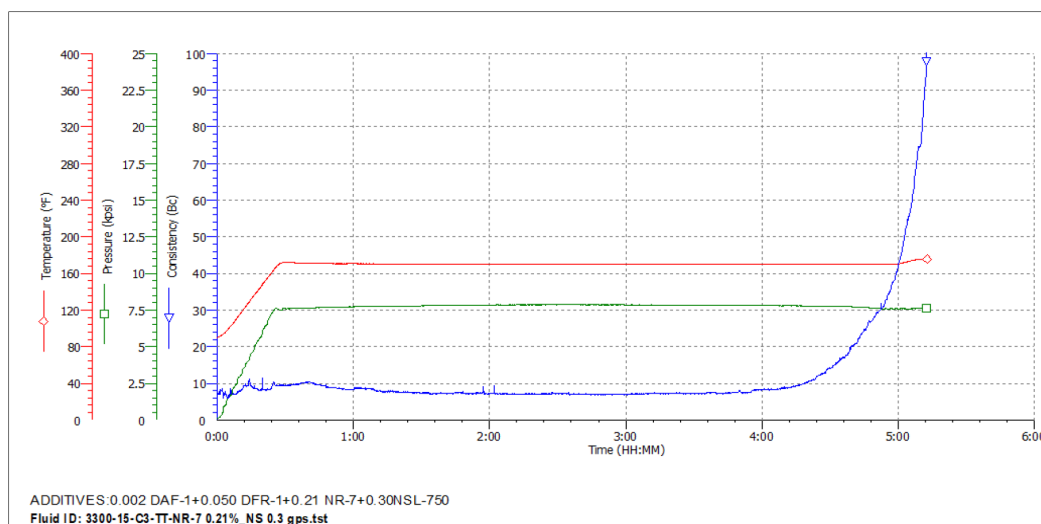


Figure 7—T.T Chart- Slurry design 0.05 % DFR-1, 0.21 % R-7, 0.3 gal/sk nano-silica

Thickening Time Consistometer

As per API standards, this machine is designed for downhole slurry setting observations. High-temperature, high-pressure atmospheric densitometer is shown in Figure 8. The thickening time test is intended to determine the time allotment the bond cement remains pumpable, a thin slurry under the bottom hole circulating temperature. The cement slurry is assessed in high high-pressure and high-temperature consistometer, which estimates the thickness of the cement slurry finalized well parameters. The machine's capacity can be tested from 600 °F and 39,000 psi (314 °C and 281 MPa) for high-temperature oil and gas wells. Consistency of liquid is estimated in Bearden units (bc).

Thickening time graph plots are divided into Temperature (°F), Pressure (psi) Consistency (Bc). The slurry was tested at a bottom hole circulating temperature (BHCT) of 170 °F; the pressure applied is 7500 (psi) as shown in the t.t. graph, starting consistency of 10- 15 Bc, gradually the slurry reaches 100 Bc slurry set.

These machines are high-temperature, pressurized densitometers to test the thickening time for cement slurry as per the suitable parameters. It is called an HPHT densitometer. The HPHT densitometer is utilized for Batch mixing and Hesitation squeeze.

Batch Mixing. The cement slurry is stirred and heated up under the bottom hole, circulating the temperature for one hour. While reporting thickening time, batch mixing is not countable.

Hesitation Squeeze

The slurry is pumped through a hesitation to reach the required squeezing pressure.

Discussions

Nano-silica blend plays a significant role and improves its properties. Various methods of preparing NS particles have been discussed in the book chapter based on previous investigations. Nano-silica, when blended in with slurry, acts as a nucleation site along these lines, quickening the hydration of cement slurry, in this way expanding its bundling thickness, which means diminishing its permeability accordingly and improving its compressive strength. Subsequently, nano-silica not only upgrades early compressive strength advancement by shortening the hydration time but also improves the compressive strength of cement slurry. Nano silica has indicated the synergic effect with retarder-07, whereby early strength improvement is seen with nano-silica focuses as a low dosage of 0.3 gallons per sack with little effect on thickening time. It has a significant impact on strength deterioration by using the Nano silica combination. Permeability is how fast water flows through a porous rock, the property of substances, openings that permit liquids or gases to pass through a permeable membrane permeable limestone. Strength retrogression will occur when the cement is exposed to elevated temperatures and with lower compressive strength and higher permeability; this is due to the formation of lime-rich crystalline phases, such as alpha-dicalcium-silicate hydrate, which are known to weaken the mechanical strength of cement and cement-based materials. Usually, strength retrogression occurs at temperatures greater than 230°F. In these cases, we must design cement slurry for high-temperature geothermal applications that typically include 33.3 to 40% additional crystalline silica to help prevent loss of compressive strength and an increase in permeability. Additionally, silica was required to provide a high-strength, stable crystalline structure. In this research, 40% additional silica can be inadequate to provide a high-strength, low-permeability cement at temperatures typically found in geothermal-well conditions of around 500°F or higher. It indicates that more considerable term strength stability in cement is typically used in geothermal-well applications.

Conclusion

The Primary motivation in this research is the Nanosilica compressive strength advancement by shortening the acceptance time of hydration, yet it also works on enhancing the compressive strength of cement. These

nano silica particles have shown the synergic impact with R-7 retarder, by which early strength improvement is seen with nano silica at low fixations.

Mixing the nano silica shows great properties when blended with slurry for at least 20 s, rheology readings approach the required flow, predictable thickening time, and great early compressive strength, which are portrayed in a referenced figures. So, by setting up this kind of concrete fittings, it tends to be utilized in a derail, for starting activity and as misfortune dissemination plug in an oil and gas tasks which is an enhanced procedure., these activities should be possible significantly quicker subsequently consequently diminishing apparatus cost and setting time, hence, this method is cost and time successful.

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